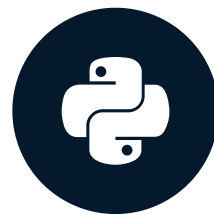


Looping through Python data structures

PYTHON FOR MATLAB USERS



Justin Kiggins
Product Manager

How for loops work in Python

MATLAB

```
for i=1:5
    disp(i);
end
disp('done');
```

```
1
2
3
4
5
done
```

Python

```
for i in [1, 2, 3, 4, 5]:
    print(i)
print('done')
```

```
1
2
3
4
5
done
```

Looping through a list of strings

```
groceries = ['bread', 'tea', 'banana']  
  
print(groceries)
```

```
['bread', 'tea', 'banana']
```

```
for item in groceries:  
    print(item)
```

```
bread  
tea  
banana
```

Looping through a dictionary

```
abbreviations = {'New York': 'NY',  
                 'California': 'CA',  
                 'Illinois': 'IL',  
                 'Texas': 'TX'}  
  
print(abbreviations)
```

```
{'New York': 'NY', 'California': 'CA', 'Illinois': 'IL', 'Texas': 'TX'}
```

```
for state, abbv in abbreviations.items():  
    print(state, abbv)
```

```
New York NY  
California CA  
Illinois IL  
Texas TX
```

Looping through a 1D NumPy array

```
import numpy as np
arr = np.array([1, 2, 3])
print(arr)
```

```
[1 2 3]
```

```
for element in arr:
    print(element)
```

```
1
2
3
```

Looping through a 2D NumPy array

```
import numpy as np
X = np.array([[1, 2, 3], [4, 5, 6]])
print(X)
```

```
[[1 2 3]
 [4 5 6]]
```

```
for row in X:
    print(row)
```

```
[1 2 3]
[4 5 6]
```

```
import pandas as pd
d = [{'fruit': 'apple', 'color': 'red'},
     {'fruit': 'banana', 'color': 'yellow'},
     {'fruit': 'pear', 'color': 'green'}]
df = pd.DataFrame(d)
print(df)
```

```
   color  fruit
0    red  apple
1  yellow banana
2   green   pear
```

```
for ii, row in df.iterrows():
    print(ii, row['fruit'], row['color'])
```

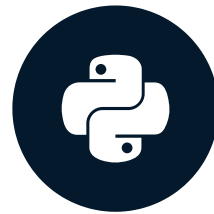
```
0 apple red
1 banana yellow
2 pear green
```


Let's practice!

PYTHON FOR MATLAB USERS

Comparison operators

PYTHON FOR MATLAB USERS



Justin Kiggins
Product Manager

Using comparison operators

```
value = 0.967  
threshold = 0.85  
  
meets_criteria = value > threshold  
  
print(meets_criteria)
```

True

The comparison operators

Comparison	Python	MATLAB
Equal	<code>==</code>	<code>==</code>
Not equal	<code>!=</code>	<code>~=</code>
Less than	<code><</code>	<code><</code>
Less than or equal	<code><=</code>	<code><=</code>
Greater than	<code>></code>	<code>></code>
Greater than or equal	<code>>=</code>	<code>>=</code>

If

```
value = 0.967
threshold = 0.85

meets_criteria = value > threshold
if meets_criteria:
    print('PASS')
```

PASS

Else

```
value = 0.275
threshold = 0.85

meets_criteria = value > threshold
if meets_criteria:
    print('PASS')
else:
    print('FAIL')
```

FAIL

Else if

```
porridge_temperature = 74.6
if porridge_temperature > 130:
    print('Too hot! :(')
elif porridge_temperature < 110:
    print('Too cold! :(')
else:
    print('Just right :D')
```

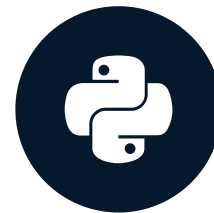
```
Too cold! :(
```

Let's practice

PYTHON FOR MATLAB USERS

Filtering data

PYTHON FOR MATLAB USERS



Justin Kiggins
Product Manager

Comparison operators and NumPy arrays

```
data = np.array([0.967, 0.56, 0.171, 0.872])  
threshold = 0.85  
meets_criteria = data > threshold  
print(meets_criteria)
```

```
[True, False, False, True]
```

Filtering NumPy arrays

```
data = np.array([-1, 0.56, -1, 0.872, 1.26])  
is_valid = data >= 0  
valid_data = data[is_valid]  
print(valid_data)
```

```
[0.56, 0.872, 1.26]
```

Filtering DataFrames

```
monkeys = df['animal'] == 'monkey'
bears = df['animal'] == 'bear'
monkey_weight = df[monkeys]['weight'].mean()
bear_weight = df[bears]['weight'].mean()
print(monkey_weight)
```

```
35.0
```

```
print(bear_weight)
```

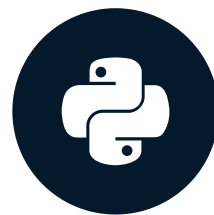
```
800.0
```

Let's practice

PYTHON FOR MATLAB USERS

Well done!

PYTHON FOR MATLAB USERS



Justin Kiggins
Product Manager

More courses to explore

Intermediate Python

Introduction to Data Visualization with Matplotlib

Introduction to Data Visualization with Seaborn

Data Manipulation with pandas

Supervised Learning with scikit-learn

Have fun!

PYTHON FOR MATLAB USERS